

(1)

If $y = b^x$, then $y' = b^x \ln b$

In general, if $y = b^{g(x)}$, then $y' = b^{g(x)} \cdot g'(x) \ln b$

Example: Differentiate the following.

a) $y = 4^x$

$$y' = 4^x \cdot \ln 4$$

$$y' = (\ln 4) 4^x$$

b) $y = 3^{x^2-4x}$

$$y' = 3^{x^2-4x} \cdot (2x-4) \ln 3$$

$$y' = 2(x-2)(\ln 3) \cdot 3^{x^2-4x}$$

c) $y = 5^{x^3+1} \cdot \sin(e^x)$

$$y' = 5^{x^3+1} \cdot (3x^2)(\ln 5) \sin(e^x) + 5^{x^3+1} \cdot \cos(e^x) \cdot e^x$$

$$y' = 5^{x^3+1} (3x^2(\ln 5) \sin(e^x) + e^x \cos(e^x))$$

↑
none.

$y =$

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Derivatives of Logarithms

(2)

If $y = \log_b x$, then $y' = \frac{1}{x \cdot \ln b}$

Proof: $y = \log_b x$

$$y = \frac{\log_e x}{\log_e b}$$

$$y = \frac{\ln x}{\ln b}$$

$$\therefore y' = \frac{1}{x \cdot \ln b}$$

If $y = \log_b g(x)$, then $y' = \frac{g'(x)}{g(x) \cdot \ln b}$

Example 1: Differentiate the following.

① $y = \log_3 x$ ② $y = \log_4(x+1)$ ③ $\log_{\frac{e}{2}} x$

$$y' = \frac{1}{x \cdot \ln 3}$$

$$y' = \frac{1}{(x+1) \cdot \ln 4}$$

$$y' = \frac{1}{x \cdot \ln\left(\frac{e}{2}\right)}$$

$$y' = \frac{1}{x [\ln e - \ln 2]}$$

$$y' = \frac{1}{x [1 - \ln 2]}$$

④ $y = \log_2 [5x^2 - 11x]$

$$y' = \frac{10x - 11}{(5x^2 - 11x) \cdot \ln 2}$$

③

⑤ Find the equation of the tangent line to the curve $y = \log_5 (x^2 - 3x)$ when $x = 4$.

Solution:

$$\frac{dy}{dx} = \frac{2x - 3}{(x^2 - 3x) \cdot \ln 5}$$

$$\left. \frac{dy}{dx} \right|_{x=4} = \frac{5}{4 \ln 5}$$

$$y - y_1 = m(x - x_1)$$

$$y - \log_5 4 = \frac{5}{4 \ln 5} (x - 4)$$

$$y = \frac{\ln 4}{\ln 5} = \frac{5}{4 \ln 5} (x - 4)$$

(4)

$$(6) \quad y = \sin(x^2) \cdot \log_3(\sin(x))$$

$$y' = \cos(x^2)(2x) \log_3(\sin(x)) + \sin(x^2) \cdot \frac{\cos(x)}{\sin(x) \cdot \ln 3}$$

$$y' = 2x \cos(x^2) \log_3(\sin(x)) + \frac{\cot(x) \sin(x^2)}{\ln 3}$$

$$y' = \frac{2x \cos(x^2) \ln(\sin(x)) + \cot(x) \sin(x^2) \ln 3}{\ln 3}$$

(7) Determine the equation of the normal line to the curve $y = \log_3(x^3 - 2x)$ at $x = 2$.

Solution

$$\frac{dy}{dx} = \frac{3x^2 - 2}{(x^3 - 2x) \cdot \ln 3}$$

$$\left. \frac{dy}{dx} \right|_{x=2} = \frac{10}{4 \ln 3} \Rightarrow m_{\perp} = -\frac{4 \ln 3}{10} = -\frac{2 \ln 3}{5}$$

$$y - \log_3 4 = -\frac{2 \ln 3}{5} (x - 2)$$

$$y - \frac{\ln 4}{\ln 3} = -\frac{2 \ln 3}{5} (x - 2)$$